

Report_DEB

1. Importation des données

```
file_path <- here::here("mod/A_DEB_SAFLYw")

l_Experiments <- c(
  "Bart2019_XP1_1",
  "Bart2019_XP1_2",
  #"Bart2019_XP2",
  "Bart2019_XP3",
  "Bart2019_XP4_1",
  "Bart2019_XP4_2",
  "Bart2019_XP5",
  "Bart2020",
  #"Gollot2026_lufa"
  "Gollot2026_dens_D1",
  "Gollot2026_dens_D2",
  "Gollot2026_dens_D2b",
  "Gollot2026_dens_D5",
  "Gollot2026_dens_D7"
)

Adults_alone <- FALSE

df_data <- f_import_data_DEB(l_Experiments, Adults_alone)
```

2. Model fit

```

NbIter <- 100000
seeds <- c(C1 = 3333, C2 = 6666, C3 = 1222)

text_priors <- '
Distrib (Fm,          TruncNormal_cv,          5,          0.2, 0.01,          10);
Distrib (pAm,         TruncNormal_cv,          100,         0.2, 40,          2000);
Distrib (r_pAm_pM,    TruncNormal_cv,          0.8,          0.1, 0.1,          2);
Distrib (v,           TruncNormal_cv,         0.18505,         0.1, 0.00001,          2);
Distrib (kap,         Uniform,                  0.01,          0.99);
Distrib (Ehp,         TruncNormal_cv,          100,          0.2, 40,          300);
Distrib (E_coc,       TruncNormal_cv,          30,          0.2, 1,          300);
Distrib (rOM_ClxHorse, LogUniform,             0.00001,         0.3);
Distrib (dv,          TruncNormal_cv,          2,          0.5, 0,          10);
Distrib (w,           LogUniform,              01,          100);

# Calcul des vraisemblances
Distrib (Sigma_W, TruncNormal, 5, 1, 0.01, 100);'

text_likelihood <- 'Likelihood (Weight,          Normal, Prediction(Weight), Sigma_W);
Likelihood (Reproduction, Poisson, Prediction(Reproduction));'

OM_diff <- TRUE

# makemcsim DEBcali.model

f_In_tot(file_path, l_Experiments, Adults_alone, OM_diff, text_priors, text_likelihood, NbIter)

# ./mcsim.DEBcali DEB_C1.in

```

OM_horse and OM_soil distinct ? TRUE

Adult and alone earthworm taken into account ? FALSE

3. Results

	Fm.1.	pAm.1.	r_pAm_pM.1.	v.1.	kap.1.	Ehp.1.
Result_mode	2.5387800	111.790000	0.84543000	0.18419500	0.68519800	114.6930
2.5%	1.9874000	101.391000	0.67615735	0.15221900	0.47952800	92.2804
97.5%	4.0976900	131.230000	0.96553300	0.22532400	0.75028900	158.3603
Min	1.7517000	72.197600	0.53182300	0.12078400	0.40639400	70.1148
Max	6.6653000	146.096000	1.13723000	0.26001400	0.82984500	175.9480

Result_mean	2.8865173	116.198848	0.82327764	0.18762481	0.61861579	124.4792
Result_sd	0.5098082	7.832348	0.07390279	0.01868699	0.06922977	16.8689
	E_coc.1.	rOM_ClxFhorse.1.	dv.1.	w.1.	Sigma_W.1.	
Result_mode	5.5743900	0.0011054400	2.8681100	14.013000	0.12457500	
2.5%	4.4781900	0.0002659560	1.5148410	9.866060	0.11086100	
97.5%	8.3029100	0.0015553900	4.7653570	32.231100	0.14027315	
Min	3.3755600	0.0000120985	0.3349210	7.380130	0.09821840	
Max	28.6675000	0.0773146000	6.2337600	38.446800	7.71175000	
Result_mean	6.3195580	0.0008269411	3.1324754	17.897013	0.12473777	
Result_sd	0.9833674	0.0005531232	0.8363664	5.605338	0.05846565	

Number of observations, n = 277
 Number of parameters, k = 10
 Log-likelihood, LL = -12.73718
 BIC = 81.71454

Potential scale reduction factors:

	Point est.	Upper C.I.
dv.1.	1.04	1.13
E_coc.1.	1.02	1.04
Ehp.1.	1.04	1.14
Fm.1.	1.67	2.79
kap.1.	1.45	2.18
pAm.1.	1.02	1.05
r_pAm_pM.1.	1.00	1.02
rOM_ClxFhorse.1.	1.06	1.20
Sigma_W.1.	1.00	1.01
v.1.	1.06	1.19
w.1.	1.80	3.22

Multivariate psrf

1.5

```

mcmc_list_corr <- mcmc.list(
  lapply(mcmc_list, function(x) {
    mcmc(as.matrix(x)[-1, , drop = FALSE]) # <- reconvertir en mcmc
  })
)

```

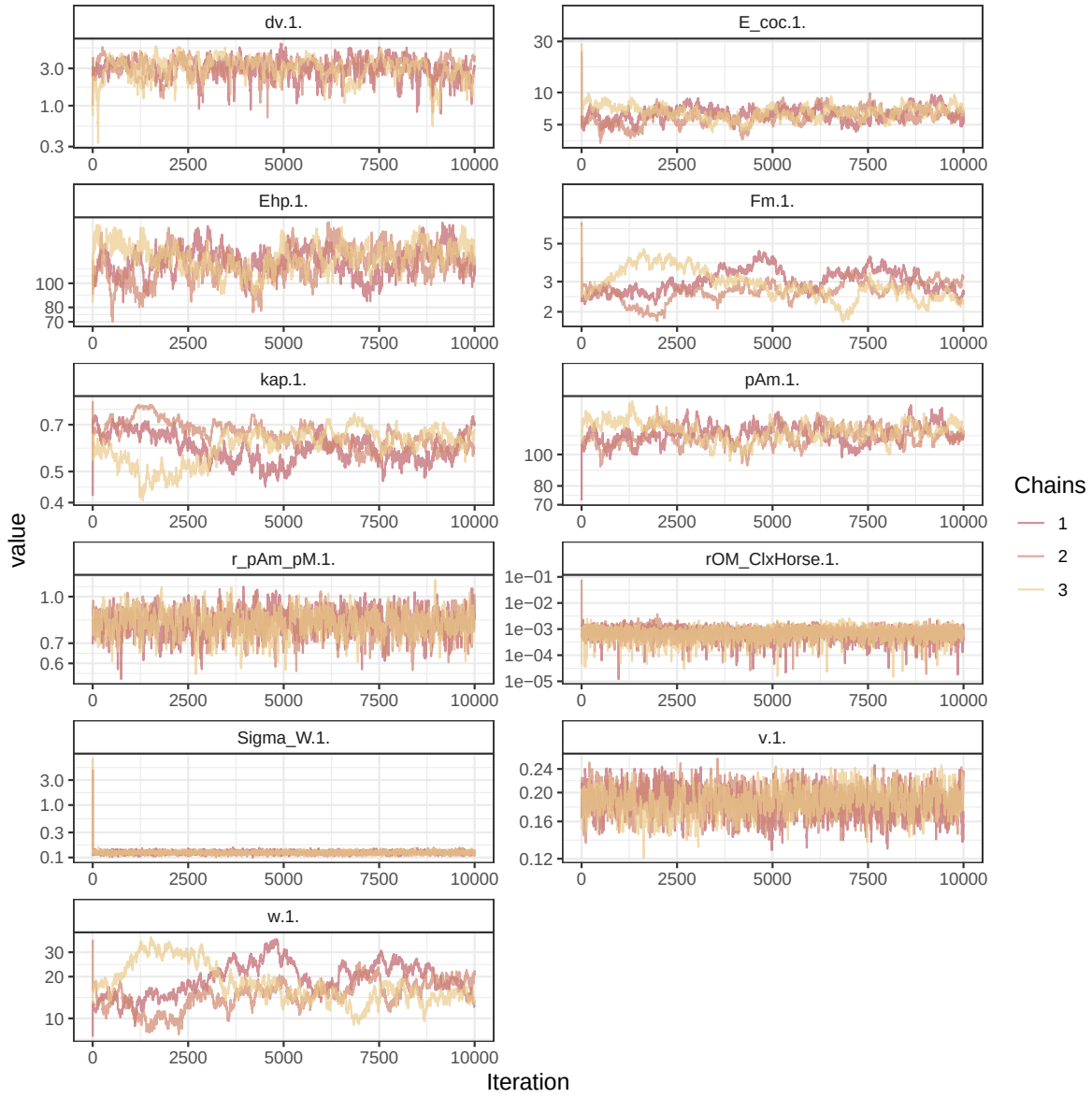


Figure 1: Parameters chains. Only the first iteration and the last r `Nb_iter_kept` are represented.

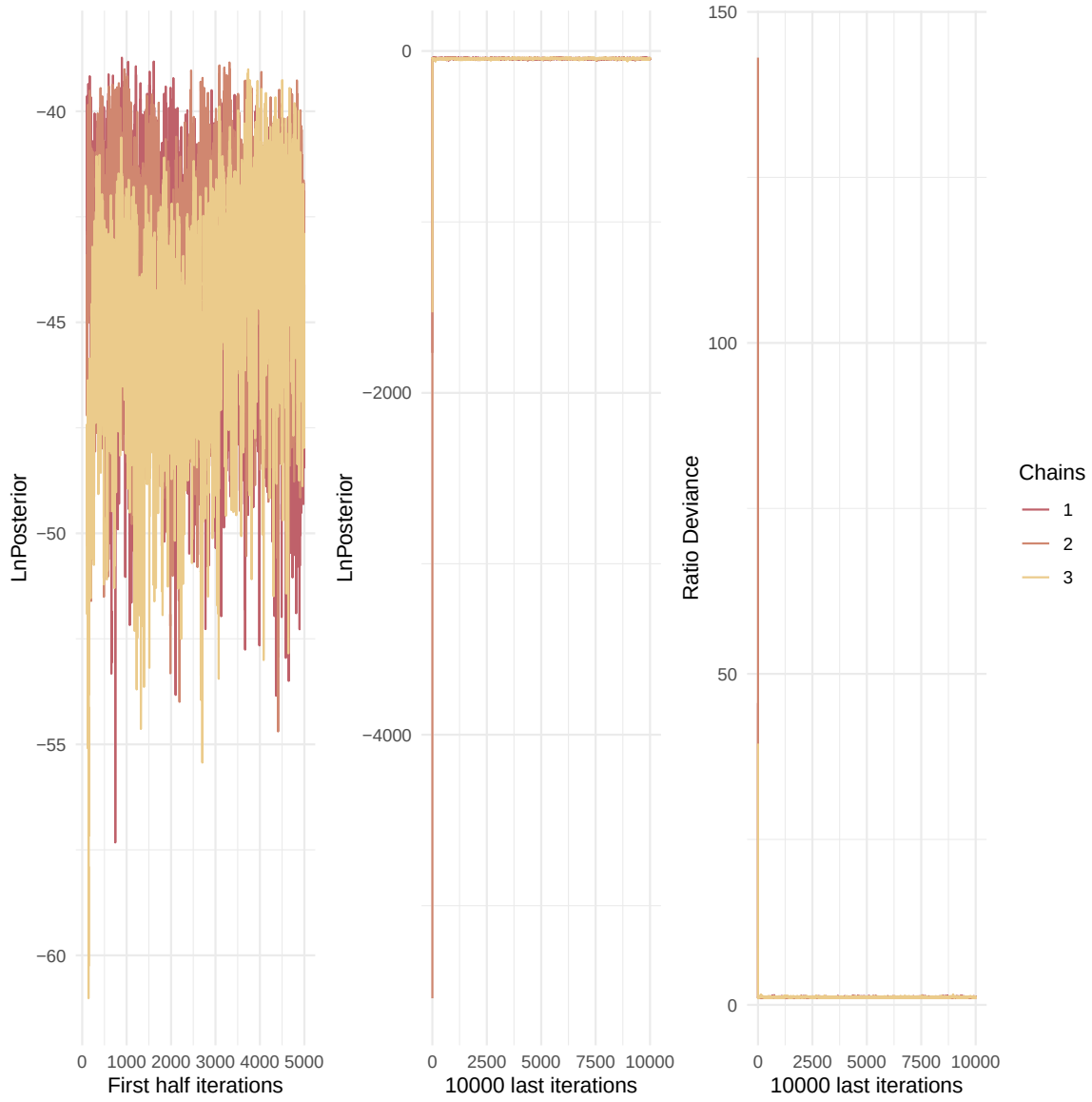


Figure 2: Evolution of likelihood and deviance ratio through the iterations.

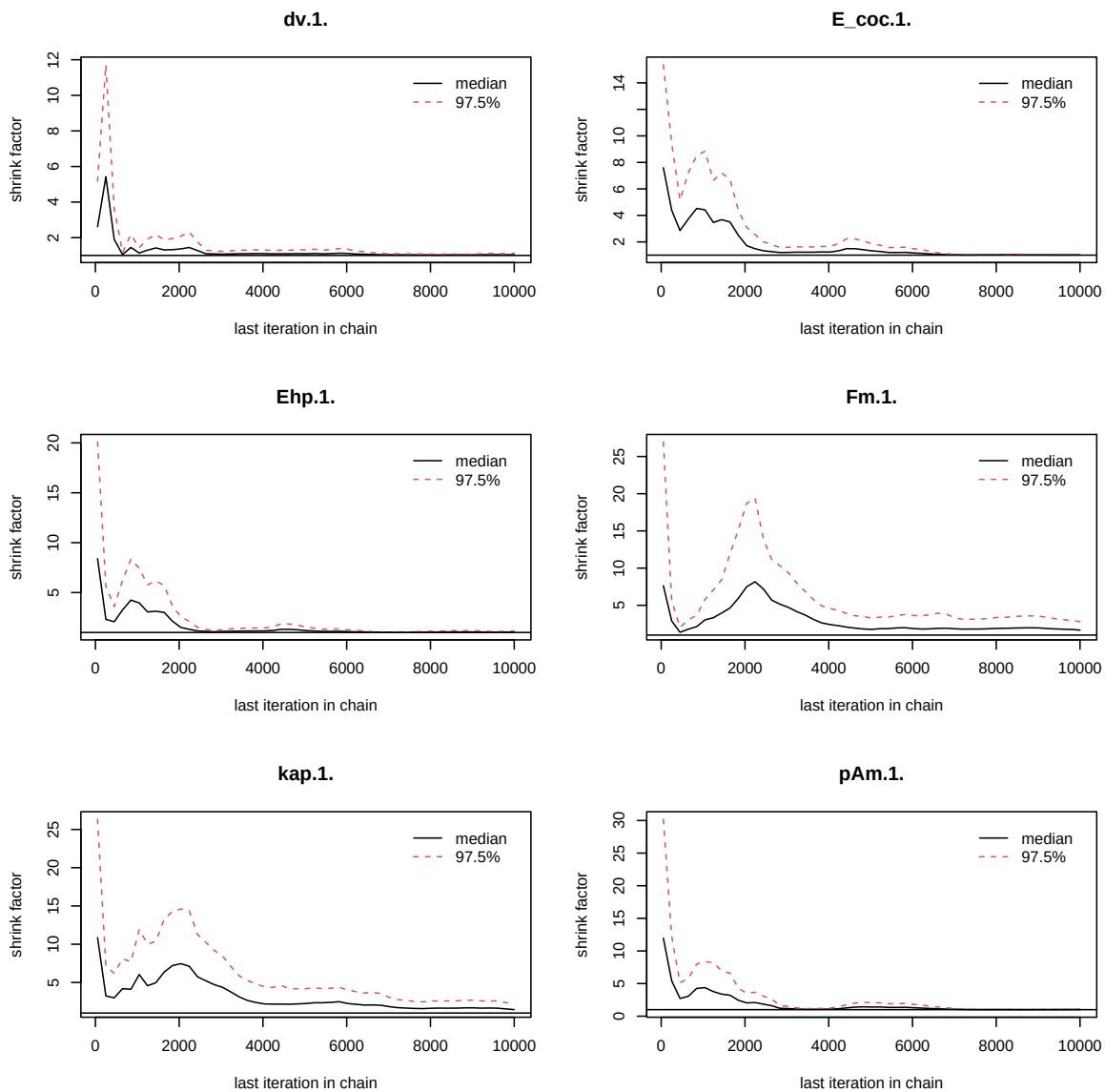


Figure 3: Chains convergence.

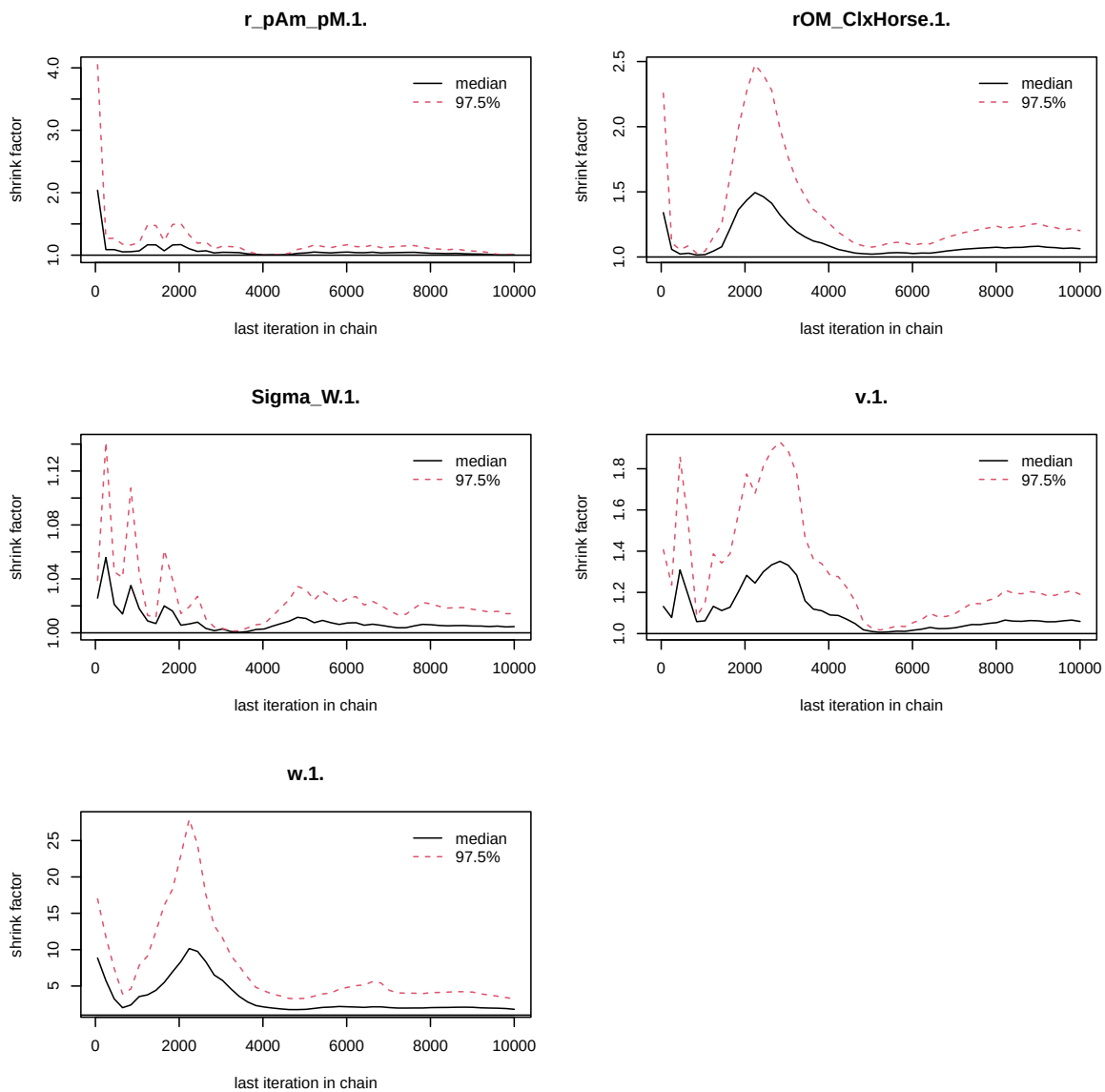


Figure 4: Chains convergence.

```
colramp_aurora <- colorRampPalette(rev(Nord_aurora[1:4]))
pairs(as.matrix(mcmc_list_corr), colramp = colramp_aurora, pixs=0.25, cex.diag = 0.5)
```

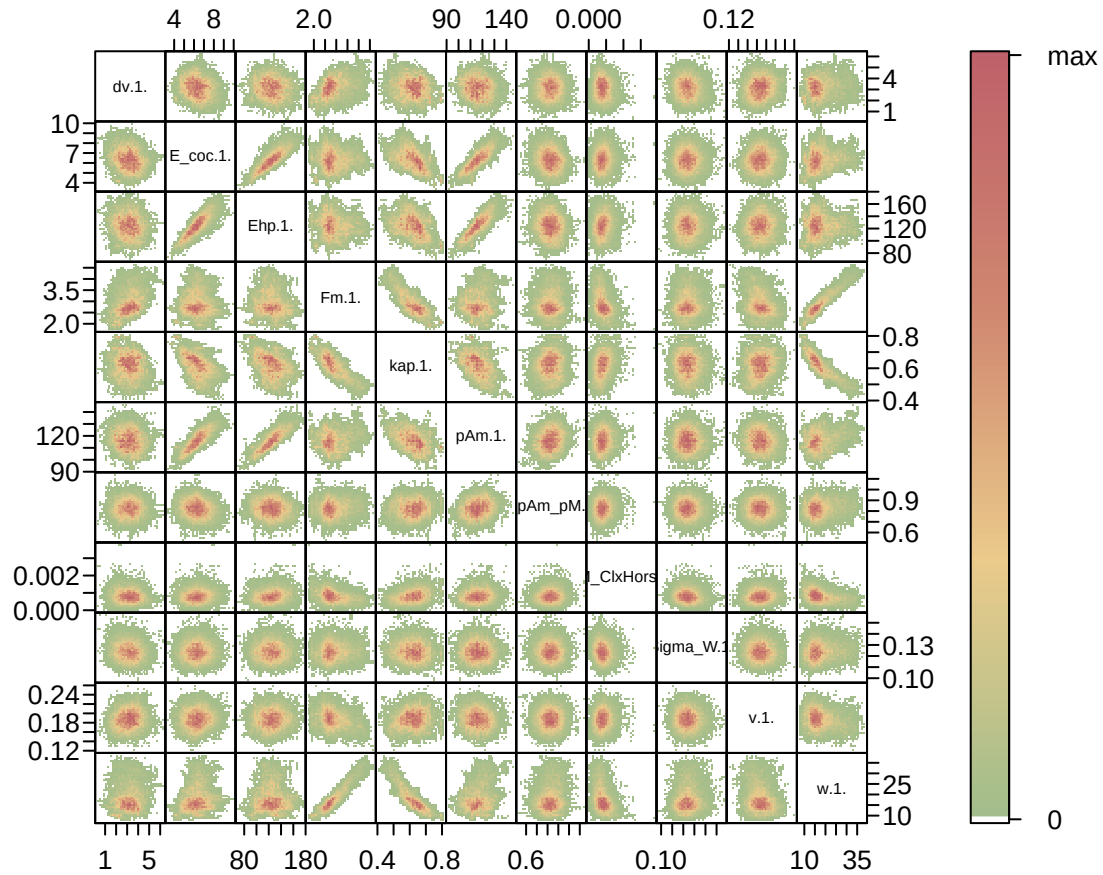


Figure 5: Parameter correlations.

4. Simulations

```
l_Experiments <- c(
  "Bart2019_XP1_1",
  "Bart2019_XP1_2",
```

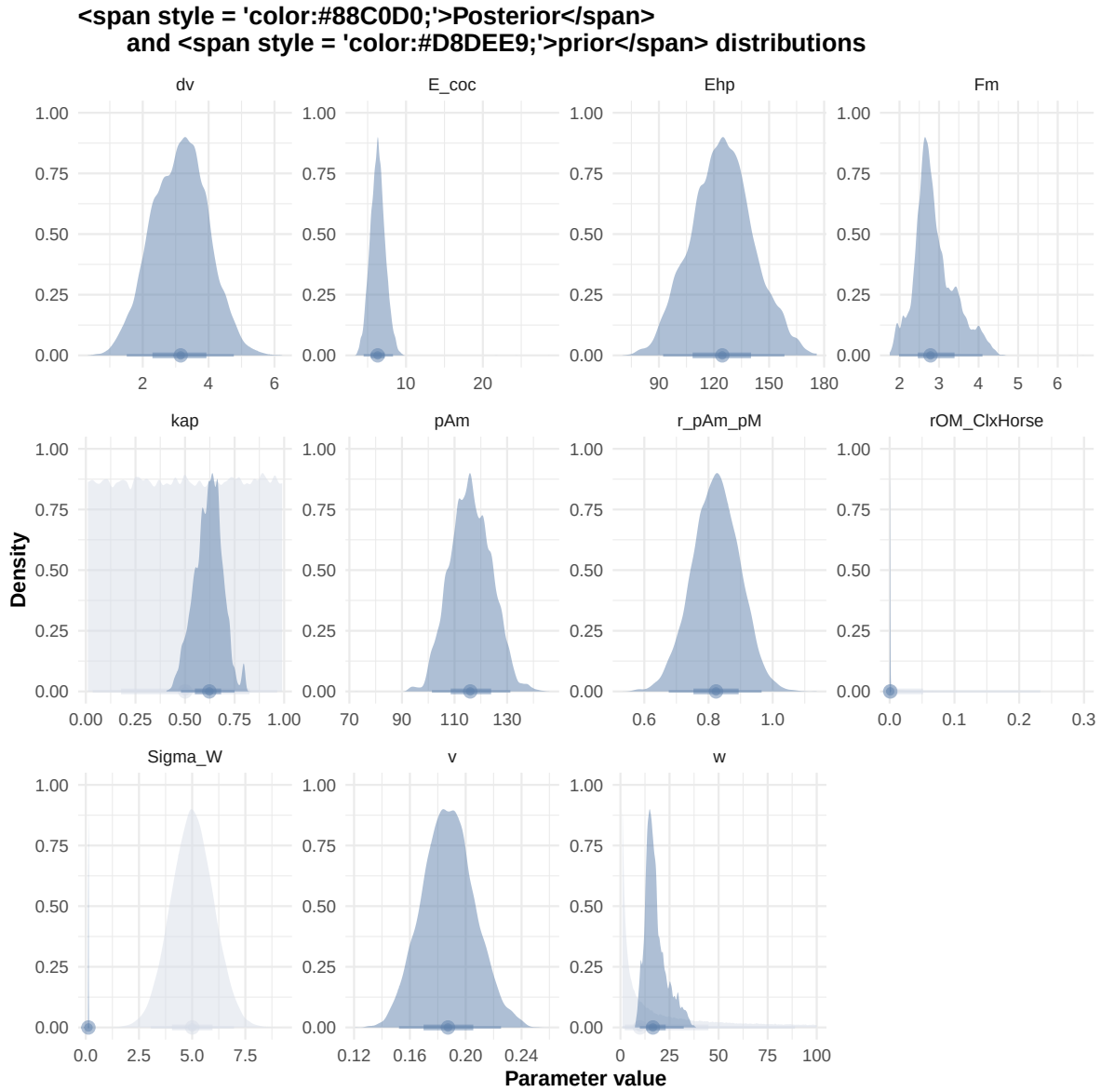


Figure 6: Distributions of the priors and the posteriors.


```
#####
df_No_sim <- df_data |>
  dplyr::select(ID_experiment, No_sim)

df_carac_color <- data.frame(
  carac = unique(l_carac),
  color = c(Nord_aurora[1], Nord_polar[2], Nord_aurora[3], Nord_aurora[2], Nord_aurora[5],
            Nord_aurora[5], Nord_frost[4], Nord_frost[3], Nord_frost[2], Nord_frost[1])
)
v_color_carac <- setNames(df_carac_color$color, df_carac_color$carac)
#####

df_data <- f_import_data_DEB(l_Experiments, Adults_alone) |>
  left_join(df_carc_exp, by="ID_experiment")

# time_limits <- df_data %>%
#   group_by(ID_experiment) %>%
#   summarise(time_cutoff = max(Time, na.rm = TRUE) + 5)

select_times <- seq(0,400, 0.1)

df_sim <- f_MCSim_read_sim(here::here(file_path, "sim.out")) |>
  left_join(df_No_sim, by="No_sim") |>
  left_join(df_carc_exp, by="ID_experiment") |>
  # left_join(time_limits, by = "ID_experiment") |>
  # filter(Time <= time_cutoff) |>
  # dplyr::select(-time_cutoff) %>%
  filter(Time %in% select_times)
```

Warning: `read\_table2()` was deprecated in readr 2.0.0.
 i Please use `read\_table()` instead.

Warning in left_join(f_MCSim_read_sim(here::here(file_path, "sim.out")), : Detected an unexpected relationship between 'x' and 'y'.
 i Row 1 of 'x' matches multiple rows in 'y'.
 i Row 1 of 'y' matches multiple rows in 'x'.
 i If a many-to-many relationship is expected, set `relationship = "many-to-many"` to silence this warning.

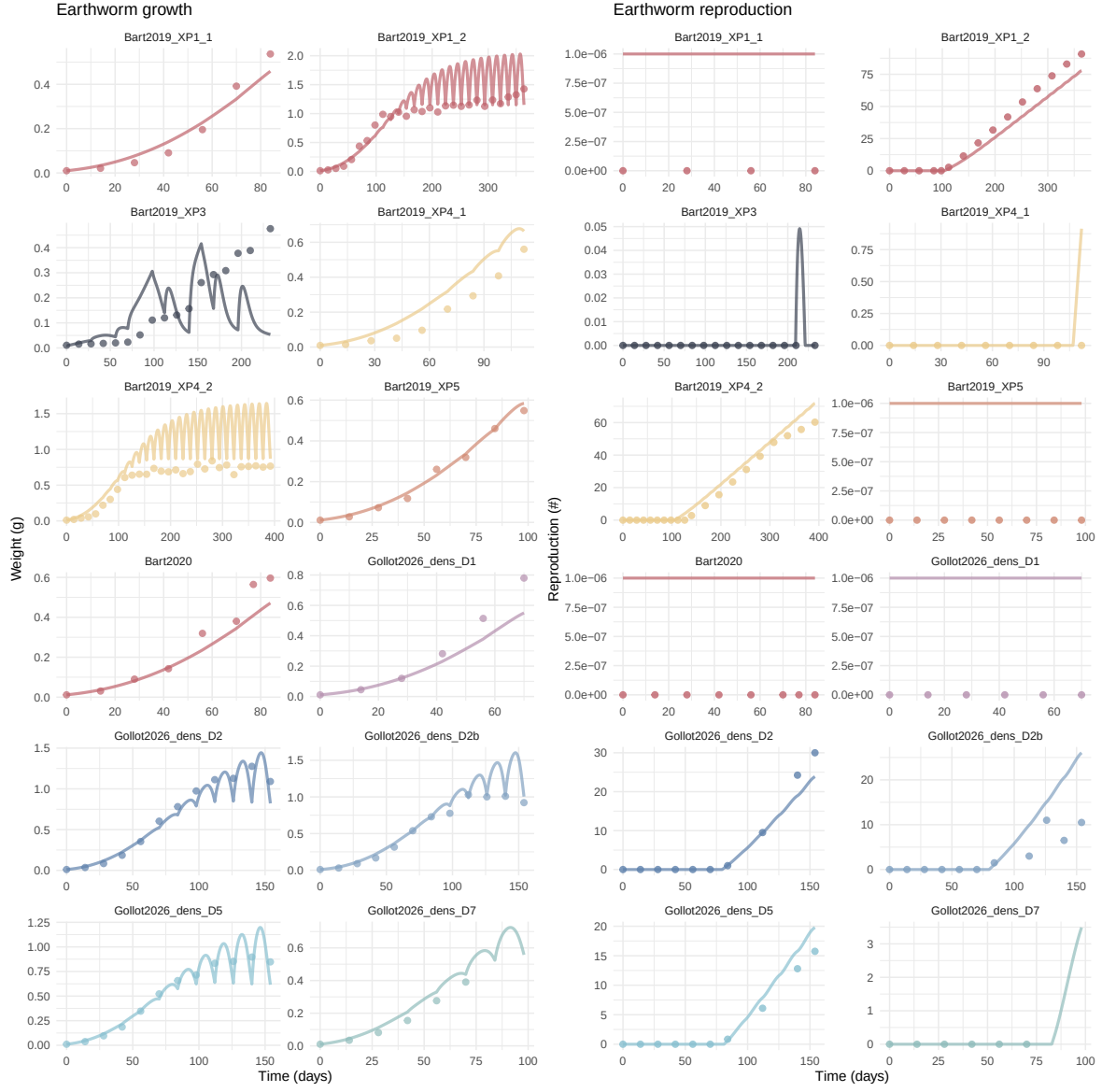
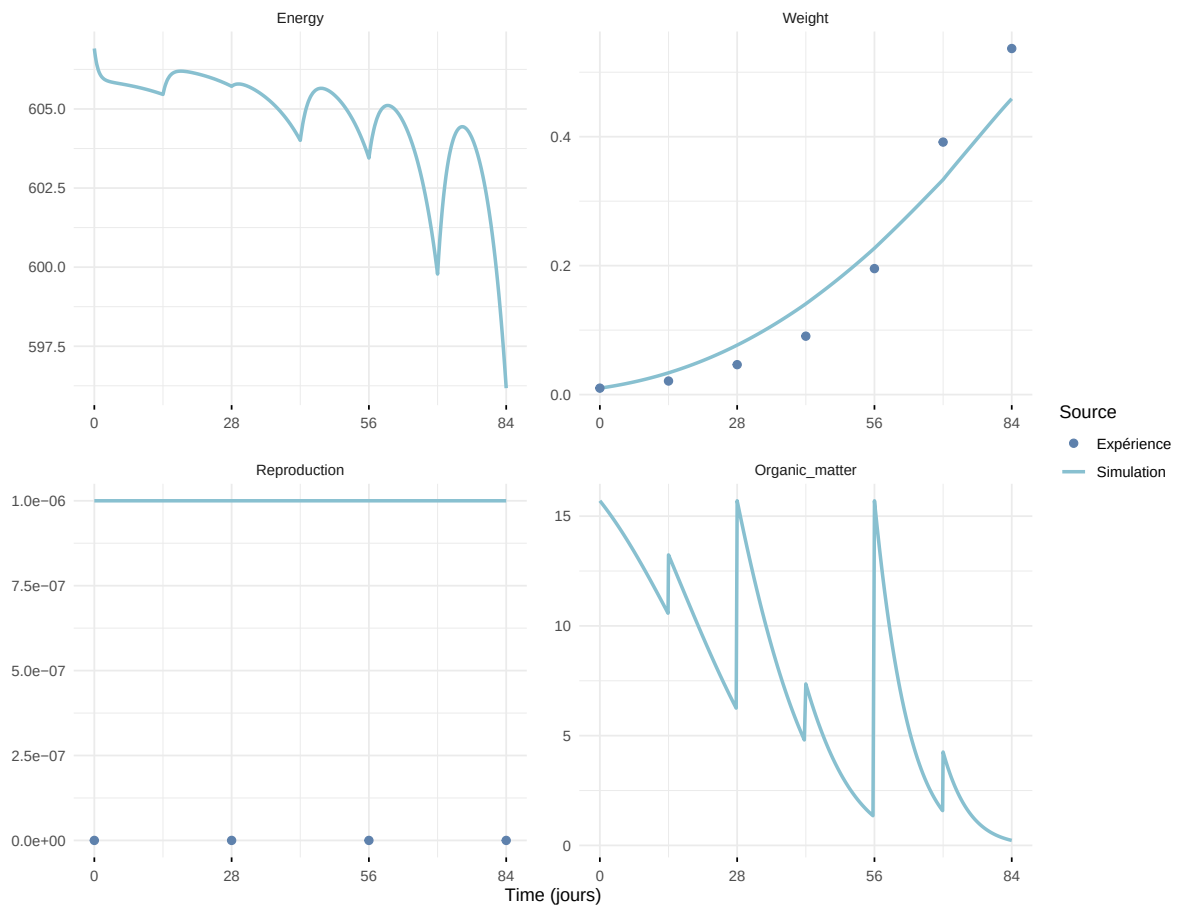
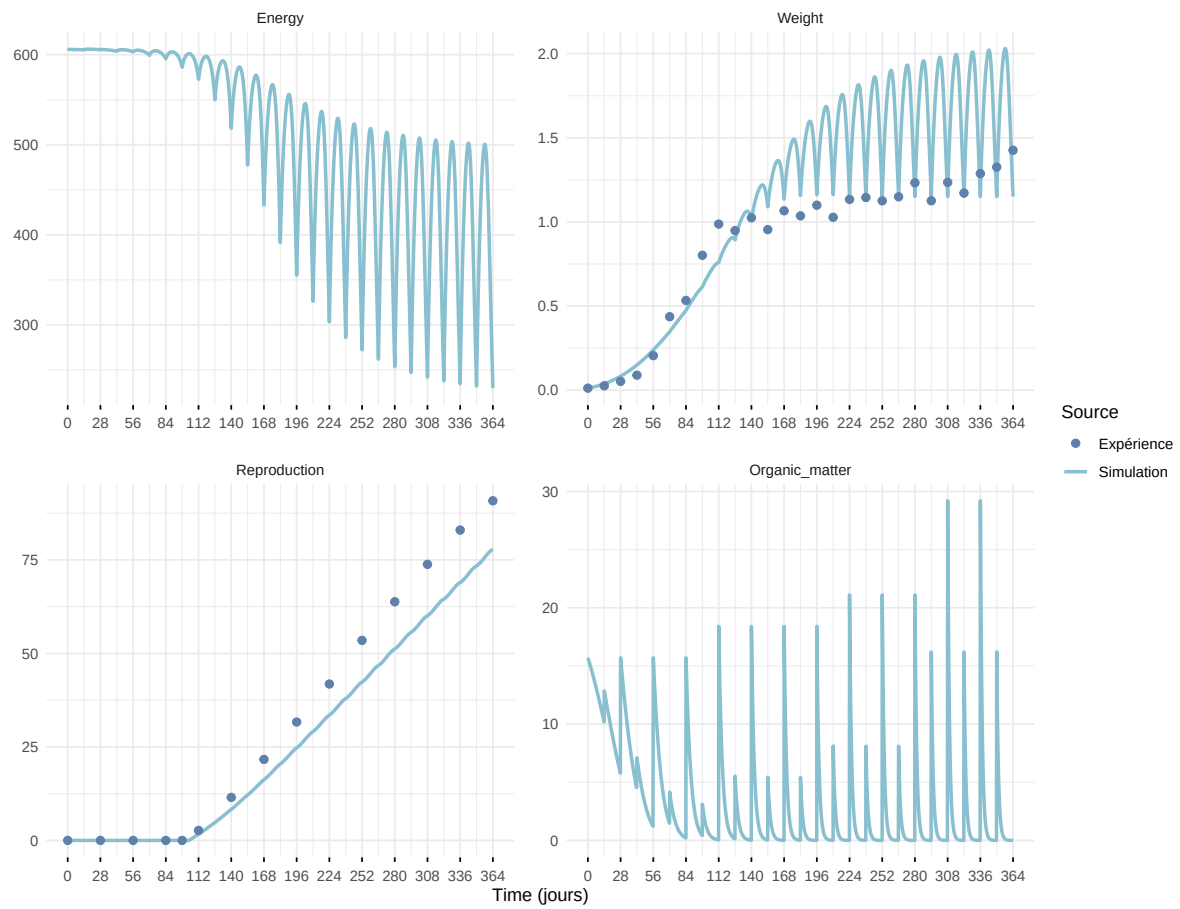


Figure 7: Simulations Weight and Reproduction.

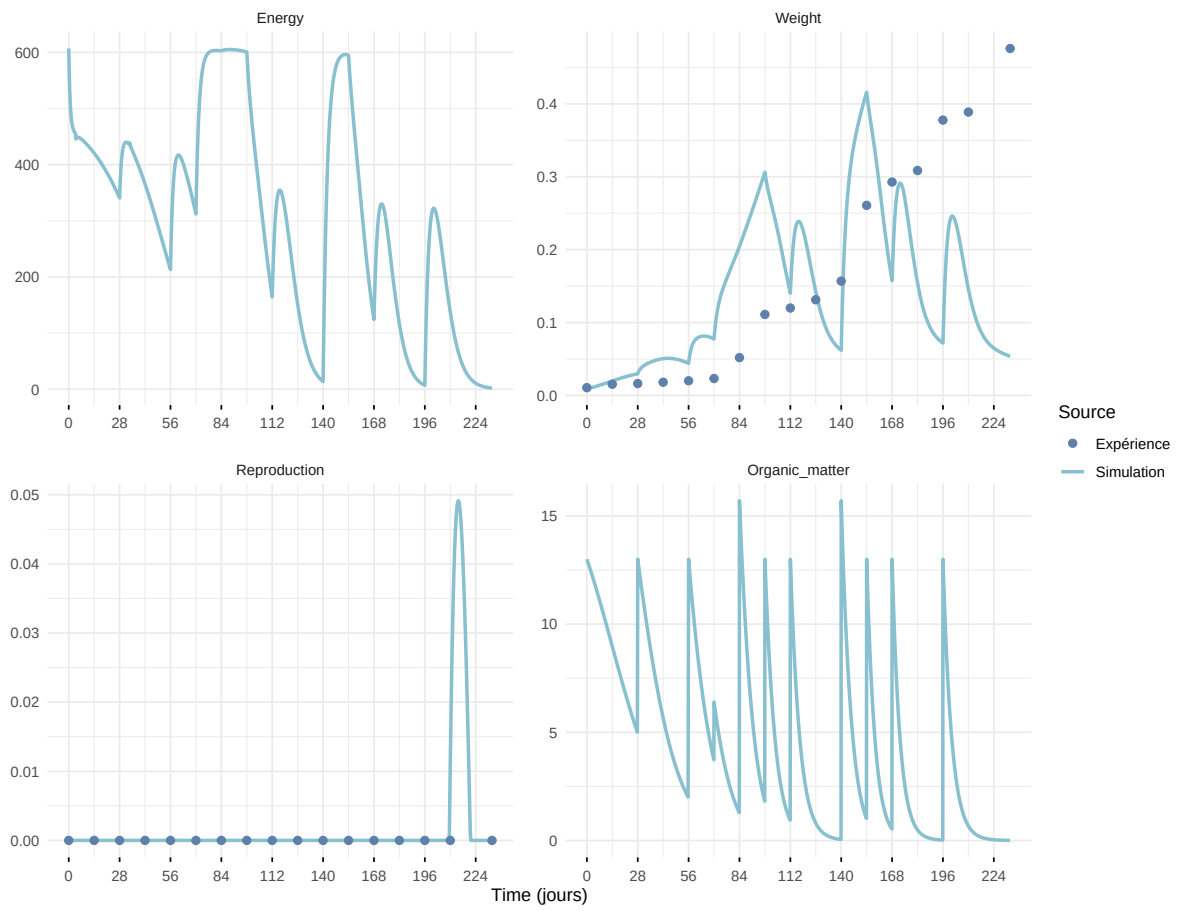
Expérience : Bart2019_XP1_1



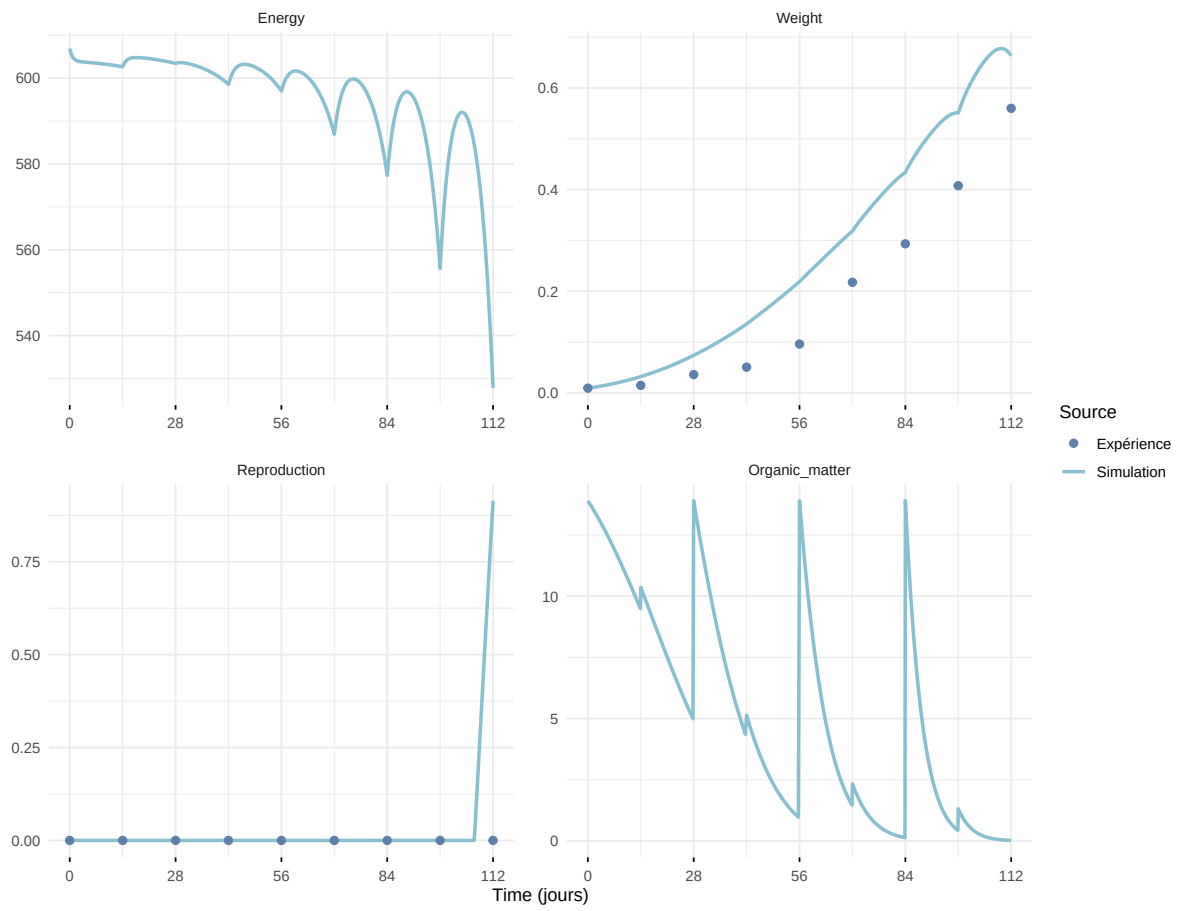
Expérience : Bart2019_XP1_2



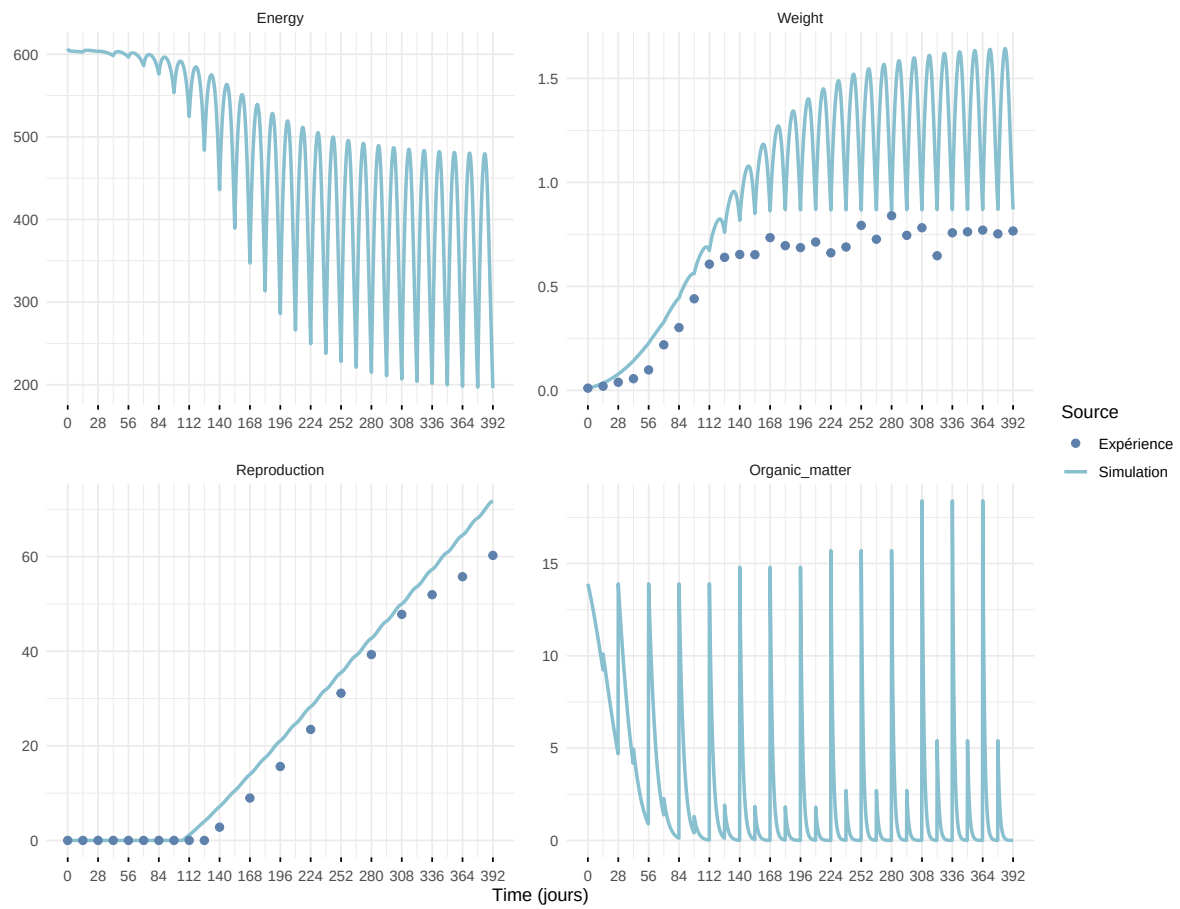
Expérience : Bart2019_XP3



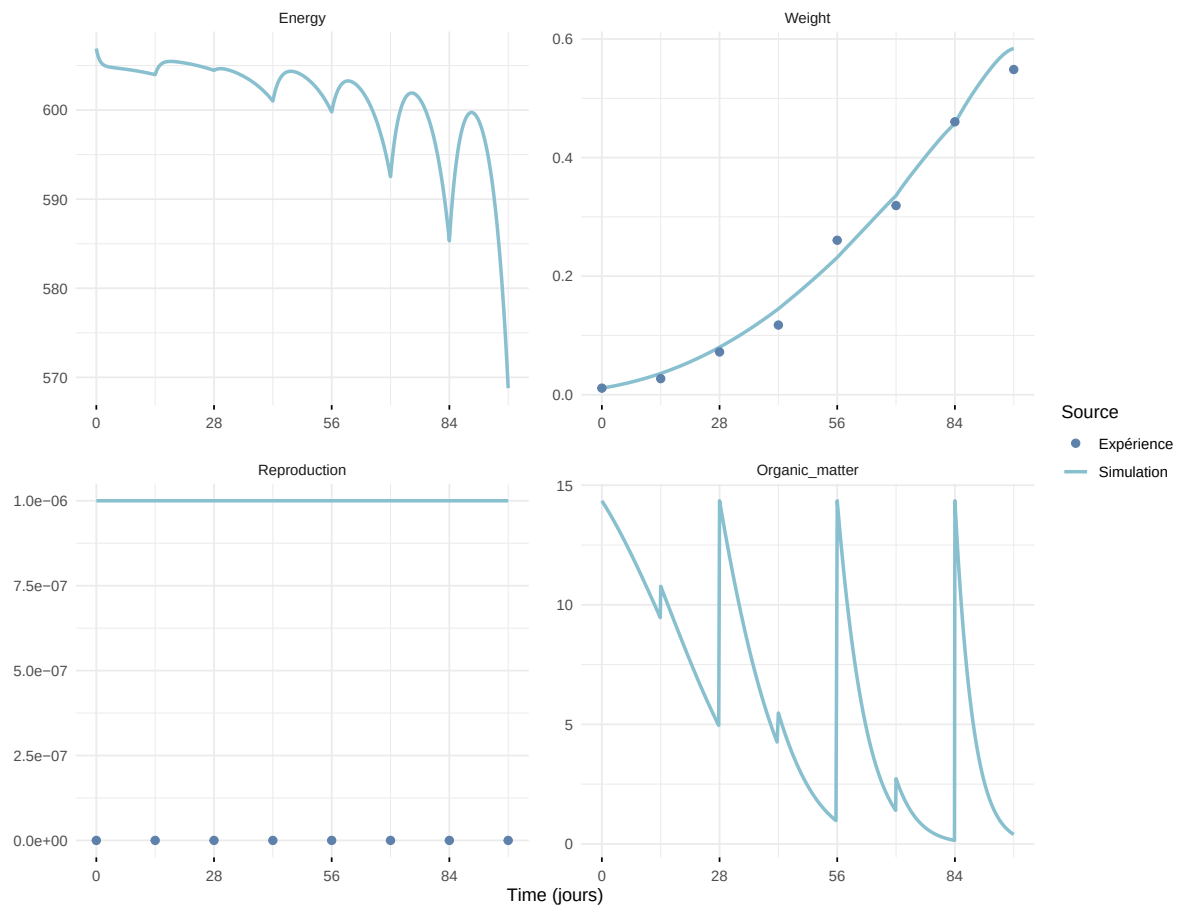
Expérience : Bart2019_XP4_1



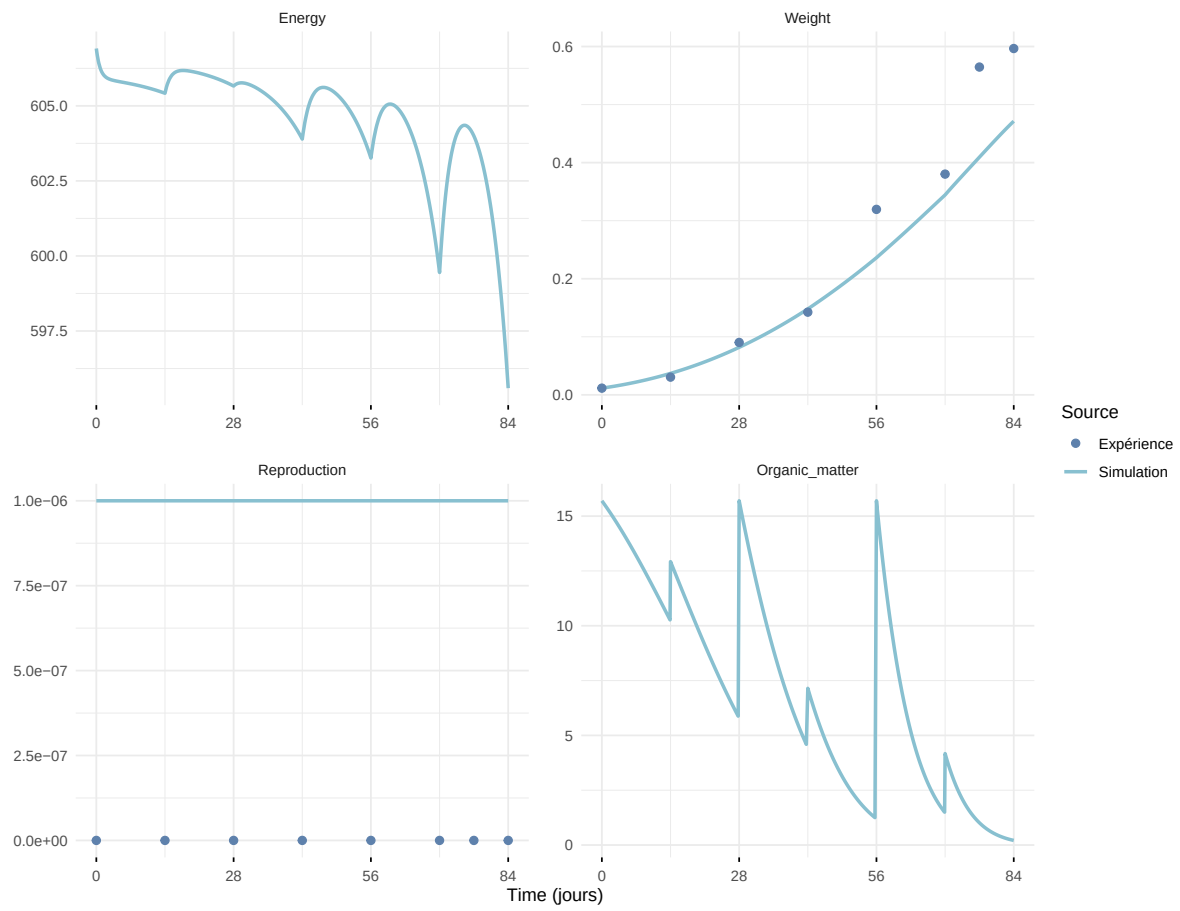
Expérience : Bart2019_XP4_2



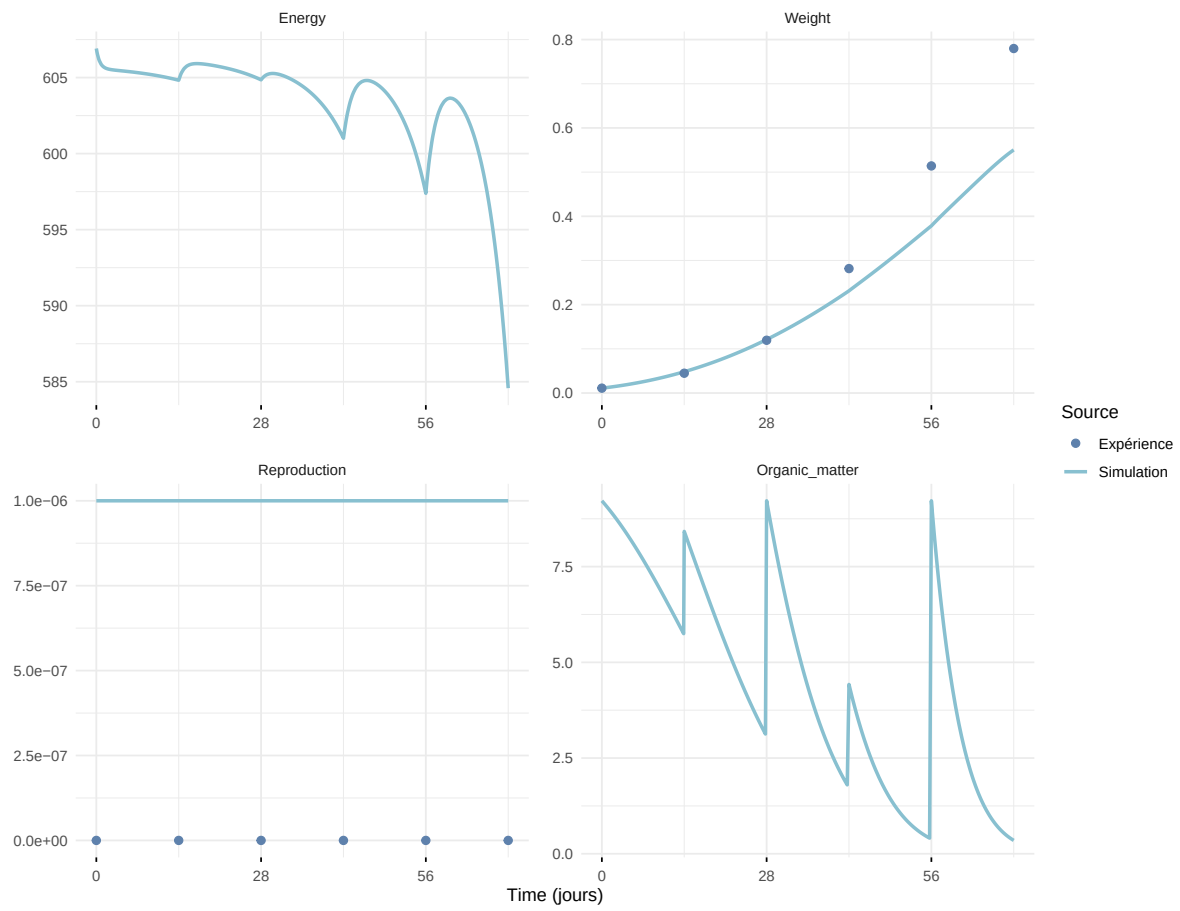
Expérience : Bart2019_XP5



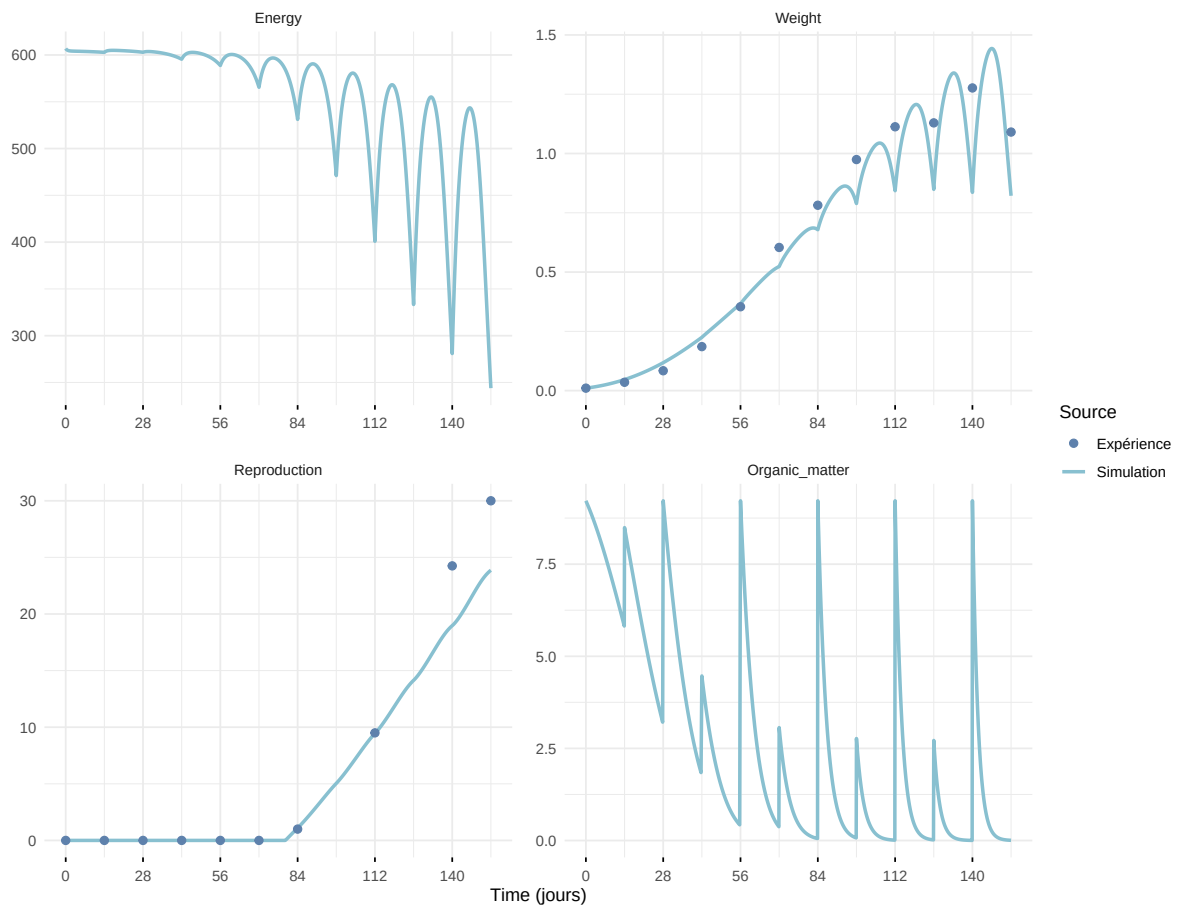
Expérience : Bart2020



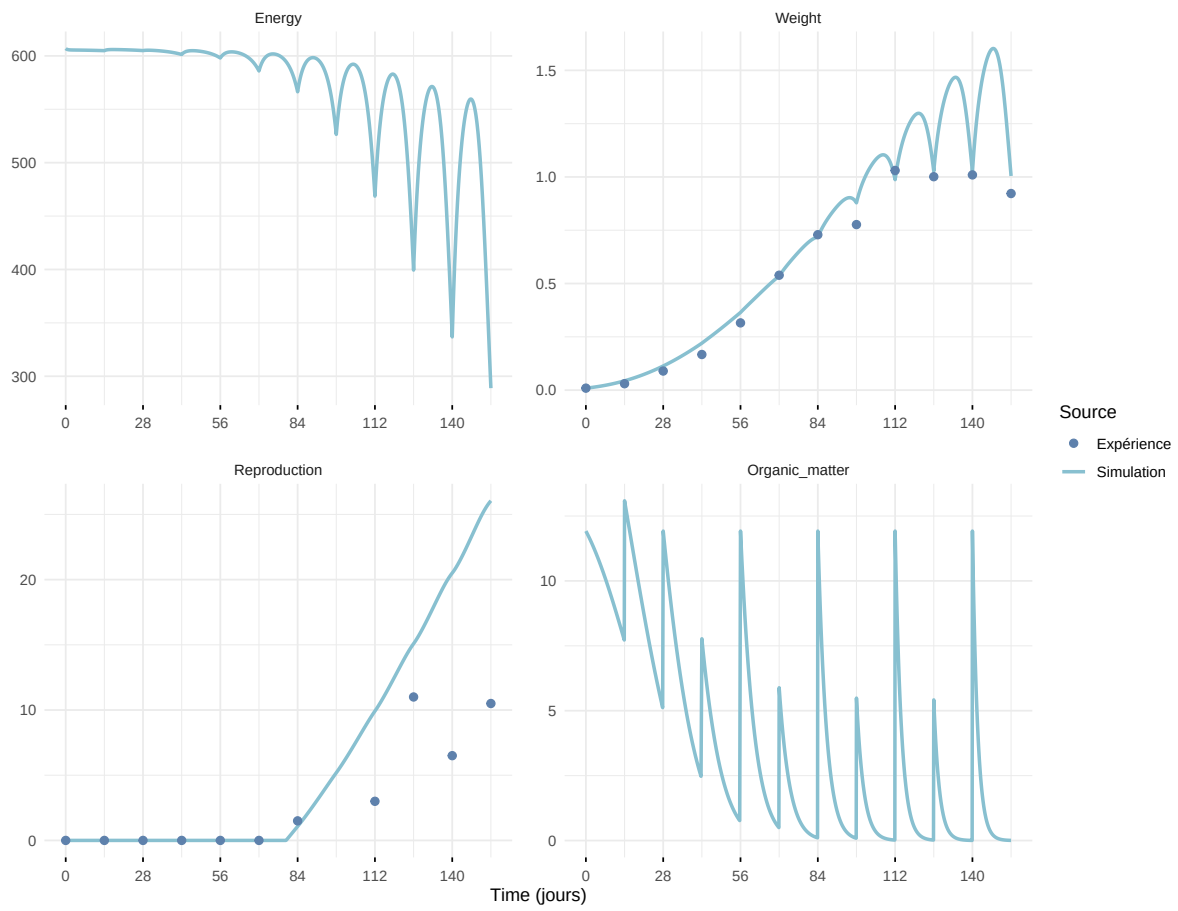
Expérience : Gollot2026_dens_D1



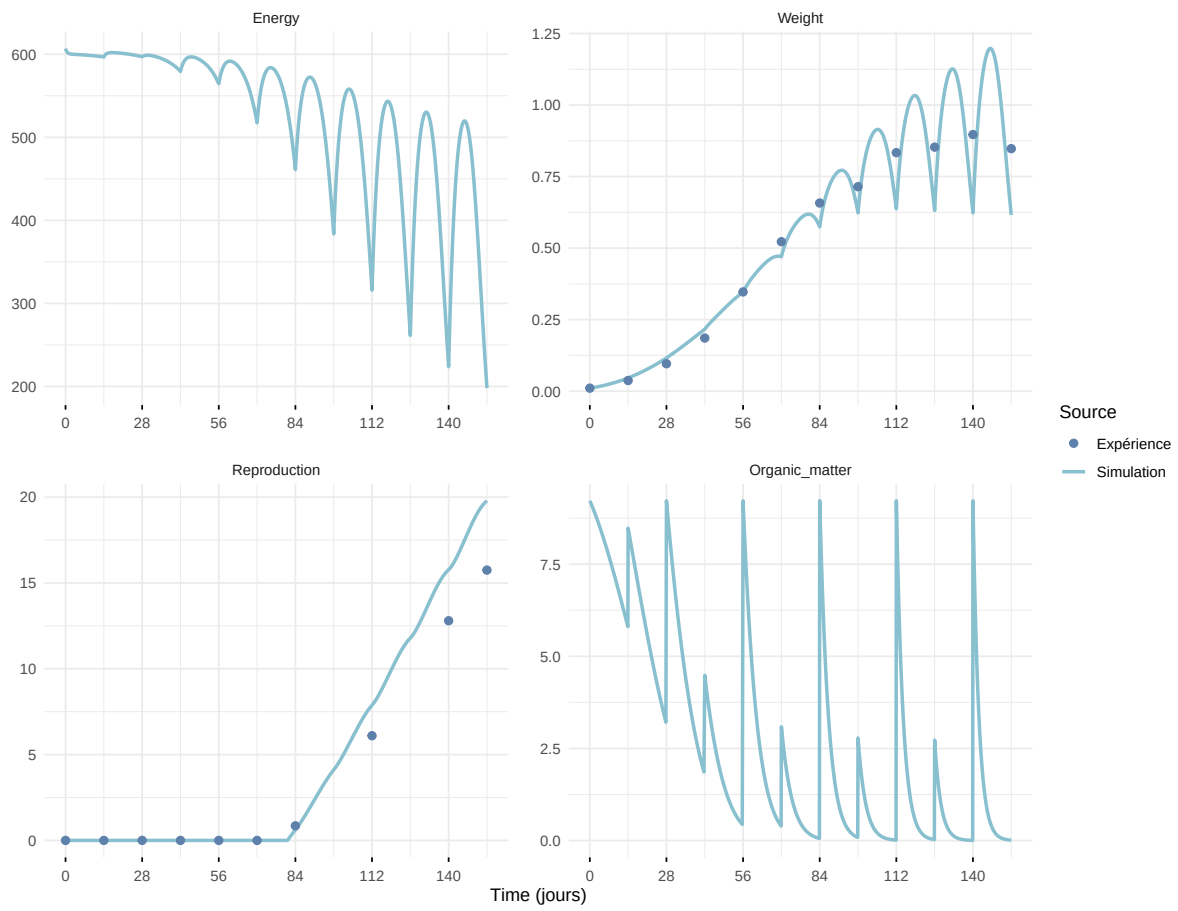
Expérience : Gollot2026_dens_D2



Expérience : Gollot2026_dens_D2b



Expérience : Gollot2026_dens_D5



Expérience : Gollot2026_dens_D7

